Paper 24 - KnightForge / N-Dimensional Chess Automata

CQE-paper-24

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Construction Basis

Generalize red/black greedy knight placement as an L-conjugate CA and then as N-dimensional chess operators.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-24.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-24\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-24\verify_knightforge_ca.py
- receipt: production\formal-papers\CQE-paper-24\knightforge_ca_receipt.json (CQE-paper-24: pass_with_open_obligations)
- body: production\papers\CQE-paper-24\PAPER-BODY.md
- source: production\papers\CQE-paper-24\SOURCE.md
- formal: production\papers\CQE-paper-24\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-24\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-24\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-24.md

Paper 24 - KnightForge / N-Dimensional Chess Automata

Abstract

Paper 24 registers greedy non-attacking knight placement as a local-rule cellular-automaton receipt whose states close under the L-conjugate centroid structure. The chessboard is the concrete shadow of the proof. The proof itself is the finite chart result: all `(L, C, R)` states anneal to one of four `L = R` attractors in at most three S3 transposition steps, and the same chart states split into the `2 + 6` VOA sector pattern.

The paper does not solve chess, N-dimensional chess, or any named OEIS sequence. It proves the CA receipt and the frame-operator lift. Playability, sequence identity, and full combinatorial-game meaning remain obligations.

Closed Evidence

The verifier calls the promoted `centroid_voa.py` substrate. The Hamming centroid verifier passes, giving four L-conjugate attractors: `(0,0,0)`, `(0,1,0)`, `(1,0,1)`, and `(1,1,1)`. Every chart state closes to that plane in at most three steps. The sector verifier passes with partition `Z(q) = 2q^0 + 6q^5`, two weight-zero vacua and six weight-five excited states. The centroid chain verifier also passes with two fixed points and six period-4 states.

The local greedy-knight verifier sweeps an 8 by 8 board in numbered boustrophedon order. At each cell it records the opposed knight-approach bits `L` and `R`, the occupancy decision `C`, the side label `R - L`, the L-conjugate attractor, the anneal step count, the three-conjugate frame label, and the VOA weight. The finite board receipt is deterministic, has occupied and rejected rows, has no attacking occupied pairs, and every row closes to an L-conjugate in at most three steps.

Definitions

A placement state is a local triple `(L, C, R)`. `C` is the current cell's occupancy decision. `L` and `R` are opposed approach bits determined by whether earlier placed knights attack the current cell from the left-approach or right-approach L-move families. A rejection is a data row, not a deletion.

An L-conjugate state is a state with `L = R`. A frame operator is a tuple of three-conjugate labels assigned to board axes or move axes. It is an operator definition, not a proof of a complete game.

Claims

- The L-conjugate attractor structure closes.

The centroid substrate proves that all eight chart states reach the four `L = R` attractors in at most three S3 transposition steps.

- The chart sectors split as `2 + 6`.

The VOA sector verifier returns `Z(q) = 2q^0 + 6q^5`: two true vacua and six excited states.

- A finite greedy knight board can be represented as a local-rule CA receipt.

The verifier sweeps a finite numbered board, emits one row per cell, records occupied and rejected decisions, and verifies that the occupied set is non-attacking.

- The board receipt inherits L-conjugate closure.

Every emitted row carries an `(L, C, R)` state and an annealing receipt. The maximum step count is at most three.

- The N-dimensional lift is a frame operator.

The verifier constructs a finite frame-operator row for four axes. It does not claim that this row is a playable, terminating, or strategically meaningful N-dimensional chess game.

Theorem 24

KnightForge is a valid CQE board-automata kernel when it returns a replayable finite placement receipt whose local states close under L-conjugate centroid annealing and whose N-dimensional extension is explicitly labeled as a frame operator rather than a solved game.

Proof

Run `verify_knightforge_ca.py`. The first three checks verify the substrate: the L-conjugate closure, the `2 + 6` sector split, and the centroid chain. These checks prove the chart structure used by the board receipt.

The fourth check constructs the finite greedy knight board. The receipt is accepted only if the sweep covers all board cells, produces both occupied and rejected rows, and leaves no occupied pair connected by a knight attack. This establishes the local-rule CA shadow of the chess example.

The fifth check applies the centroid closure to every board row. A failed anneal, or a row requiring more than three steps, would falsify the claim that the board receipt is carried by the L-conjugate chart structure.

The final check builds the N-dimensional frame operator and verifies that it is not represented as a closed game claim. Thus the paper proves the finite CA and frame-lift receipt while carrying the game-theoretic obligations.

Receipt

The formal receipt is generated at `production/formal-papers/CQE-paper-24/knightforge_ca_receipt.json`.

Open Obligations

OEIS identity remains open. N-dimensional playability remains open. The placement-class relation to the `2 + 6` sector split remains open beyond the local chart receipt. Combinatorial-game expert review remains open.

Suite Role

Paper 23 supplies the chain/path descriptor discipline. Paper 24 applies that discipline to board automata. Paper 25 may reuse the board receipt as a move-cost or traversal ledger, while Paper 28 may develop broader game lattices.